Assignment 16
System Design & Scenario Questions

Part D — Interview Scenarios

10 questions

★ **Must-know** — asked very frequently in GenAI / ML interviews (2 questions). Prepare these first.

1. Design a chatbot that answers questions over 10,000 internal company PDFs. Cover ingestion, chunking, embeddings, vector DB, retrieval, reranking, LLM choice, evaluation and monitoring. ★
2. Your RAG chatbot answers correctly in testing but users complain in production. How do you investigate? ★
3. Design a customer-support agent that can look up orders and issue refunds safely (tools, permissions, human approval).
4. Your LLM API bill has tripled this month. What would you check and change?
5. A user's question needs data from both a SQL database and documents. Design the system (Text-to-SQL + RAG routing).
6. How would you build a multilingual (e.g. Hindi/Marathi + English) GenAI assistant?
7. Design a document-extraction pipeline that pulls structured fields from invoices/forms with an LLM.
8. How would you handle confidential data when using a third-party LLM API?
9. P95 latency of your assistant is 12 seconds. How do you bring it under 3 seconds?
10. How would you A/B test two prompts or two models in production?